

Closure properties for languages

Language	Operation				
	$L_1 \cup L_2$	$L_1 \cap L_2$	$\bar{L}$	$L_1 \circ L_2$	L^*
DFA	Easy	Easy	Easy	Hard	Hard
Regex	Easy	Hard	Hard	Easy	Easy
NFA	Easy	Hard	Hard	Easy	Easy

- $L_1 \cup L_2$ = Union of L_1 and L_2
- $L_1 \cap L_2$ = Intersection of L_1 and L_2
- $\bar{L}$ = Complement of L
- $L_1 \circ L_2$ = Concatenation of L_1 and L_2
- L^* = Powers of L

Properties

Let L_1 and L_2 be regular languages.

Then, the following languages are **regular**.

- **Complement.** $\bar{L}_1 = \{x \mid x \in \Sigma^* \text{ and } x \notin L_1\}$.
- **Union.** $L_1 \cup L_2 = \{x \mid x \in L_1 \text{ or } x \in L_2\}$.
- **Intersection.** $L_1 \cap L_2 = \{x \mid x \in L_1 \text{ and } x \in L_2\}$.
- **Concatenation.** $L_1 \cdot L_2 = \{xy \mid x \in L_1 \text{ and } y \in L_2\}$.
- **Star.** $L_1^* = \{x_1x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in L_1\}$.

Lecture 4

Let the alphabet Σ be the standard 26 letters $\{a, b, \dots, z\}$. If $A = \{\text{good}, \text{bad}\}$ and $B = \{\text{boy}, \text{girl}\}$, then

$$A \cup B = \{\text{good}, \text{bad}, \text{boy}, \text{girl}\},$$

$$A \circ B = \{\text{goodboy}, \text{goodgirl}, \text{badboy}, \text{badgirl}\}, \text{ and}$$

$$A^* = \{\epsilon, \text{good}, \text{bad}, \text{goodgood}, \text{goodbad}, \text{badgood}, \text{badbad}, \\ \text{goodgoodgood}, \text{goodgoodbad}, \text{goodbadgood}, \text{goodbadbad}, \dots\}.$$

Let $\mathcal{N} = \{1, 2, 3, \dots\}$ be the set of natural numbers. When we say that $\mathcal{N}$ is *closed under multiplication*, we mean that for any x and y in $\mathcal{N}$, the product $x \times y$ also is in $\mathcal{N}$. In contrast, $\mathcal{N}$ is not closed under division, as 1 and 2 are in $\mathcal{N}$ but $1/2$ is not. Generally speaking, a collection of objects is **closed** under some operation if applying that operation to members of the collection returns an object still in the collection. We show that the collection of regular languages is closed under all three of the regular operations. In Section 1.3, we show that these are useful tools for manipulating regular languages and understanding the power of finite automata. We begin with the union operation.

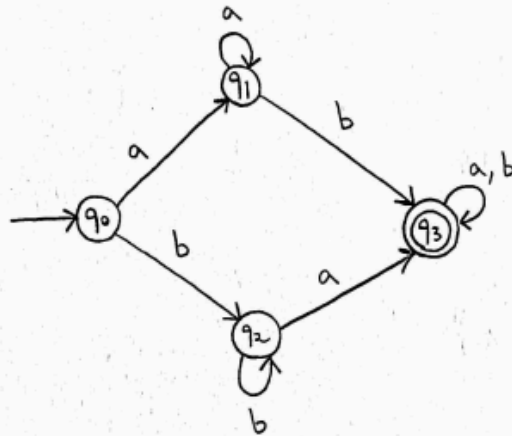
1. $L_3 = \{w \mid w \in \{0, 1\}^* \text{ } w \text{ contains neither the substring } ab \text{ nor } ba\}$

First complement the language:

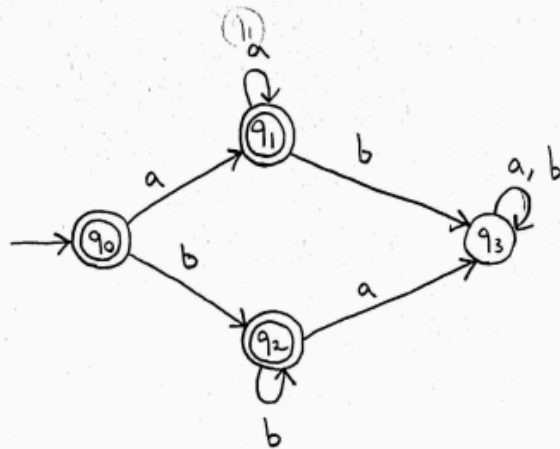
$$\overline{L} = \{w \mid w \text{ contains either the substring } ab \text{ or } ba\}$$

Lecture 4

$\Rightarrow$ DFA for $\bar{L}$:



$\Rightarrow$ DFA for L : Accept states in $\bar{L}$ become reject states in L
Reject states in $\bar{L}$ become accept states in L



Union

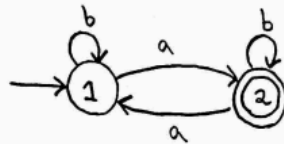
Lecture 4

$$L = \{ w \mid w \text{ has an odd number of } a\text{'s and ends with a } b \}$$

$$\Rightarrow L_1 = \{ w \mid w \text{ has an odd number of } a\text{'s} \}$$

$$\Rightarrow L_2 = \{ w \mid w \text{ ends with a } b \}$$

DFA for L_1 :



1. Set of states: $Q = \{1, 2\}$

2. Input alphabet: $\Sigma = \{a, b\}$

3. Transition function: $\delta: Q \times \Sigma \rightarrow Q$

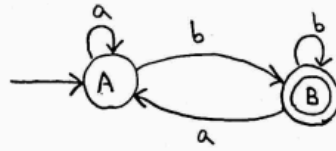
	a	b	
1	2	1	$\Rightarrow \delta(1, a) = 2, \delta(1, b) = 1$
2	1	2	$\Rightarrow \delta(2, a) = 1, \delta(2, b) = 2$

4. Start State: $q_0 = 1$

5. Set of Final States: $F = \{2\}$

Lecture 4

DFA for L_2 :



1. Set of states: $Q = \{A, B\}$
2. Input alphabet: $\Sigma = \{a, b\}$
3. Transition function: $\delta: Q \times \Sigma \rightarrow Q$

	a	b	
A	A	B	$\Rightarrow \delta(A, a) = A, \delta(A, b) = B$
B	A	B	$\Rightarrow \delta(B, a) = A, \delta(B, b) = B$

4. Start state: $q_0 = A$
5. Set of Final States: $F = \{B\}$

The two DFA L_1 and L_2 are existing separately.

Run the input **aaab** on the DFA for L_1 , the input will be consumed. This is a downside, since the DFA cannot rewind the input tape and then run the input in the DFA of L_2 . To solve this problem we need to run the DFA parallelly so both the condition for the language is satisfied. We use a technique called cross product.

Lecture 4

DFA for L using L_1 and L_2 :

1. Set of states: $Q = \{1A, 1B, 2A, 2B\}$

2. Input alphabet: $\Sigma = \{a, b\}$

3. Transition function: For each $(r_1, r_2) \in Q$ and each $a \in \Sigma$, $\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$

$$\begin{aligned}\delta((1, A), a) &= (\delta_1(1, a), \delta_2(A, a)) \\ &= (2, A)\end{aligned}$$

$$\begin{aligned}\delta((1, A), b) &= (\delta_1(1, b), \delta_2(A, b)) \\ &= (1, B)\end{aligned}$$

$$\begin{aligned}\delta((1, B), a) &= (\delta_1(1, a), \delta_2(B, a)) \\ &= (2, A)\end{aligned}$$

$$\begin{aligned}\delta((1, B), b) &= (\delta_1(1, b), \delta_2(B, b)) \\ &= (1, B)\end{aligned}$$

$$\begin{aligned}\delta((2, A), a) &= (\delta_1(2, a), \delta_2(A, a)) \\ &= (1, A)\end{aligned}$$

$$\begin{aligned}\delta((2, A), b) &= (\delta_1(2, b), \delta_2(A, b)) \\ &= (2, B)\end{aligned}$$

$$\begin{aligned}\delta((2, B), a) &= (\delta_1(2, a), \delta_2(B, a)) \\ &= (1, A)\end{aligned}$$

$$\begin{aligned}\delta((2, B), b) &= (\delta_1(2, b), \delta_2(B, b)) \\ &= (2, B)\end{aligned}$$

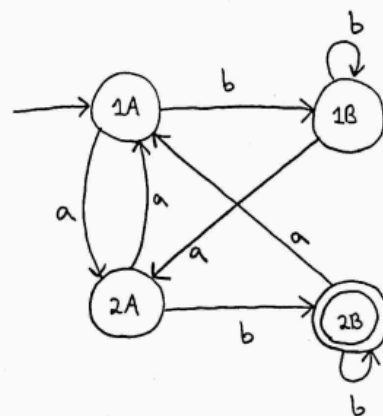
Transition Table for L :

	a	b
1A	2A	1B
1B	2A	1B
2A	1A	2B
2B	1A	2B

4. Start state: $q_0 = 1A$

5. Set of Final States: $F = \{2B\}$

DFA for L :



Lecture 4

AND in language

Set of final states $F = \{2B\}$

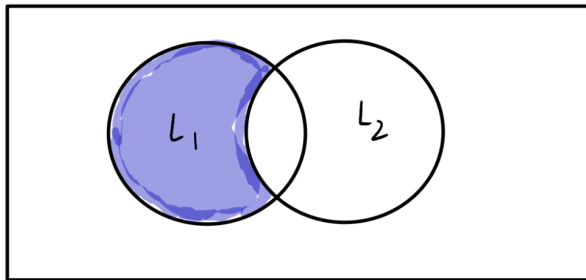
OR in language

Set of final states $F = \{1B, 2A, 2B\}$

XOR in language

Set of final states $F = \{1B, 2A\}$

What if the language said $L_1 - L_2 \rightarrow$ Difference



$$L_1 - L_2 = L_1 / L_2 \\ = L_1 \cap \bar{L}_2$$

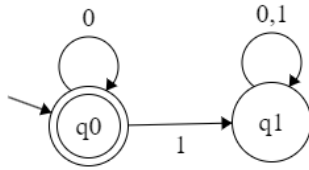
Here if we complement L_2 , the final state will be A.

$$\therefore F = \{2A\}$$

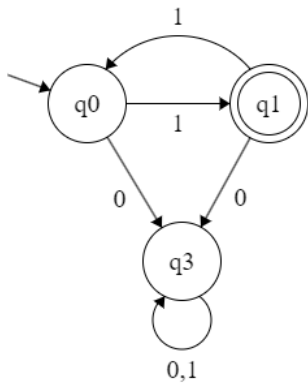
Lecture 4

Concatenation

$$L_1 = \{w \in \{0, 1\}^* \mid 0^n \text{ where } n \geq 0\}$$

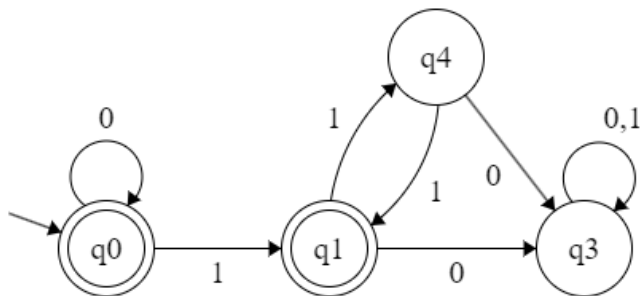


$$L_2 = \{w \in \{0, 1\}^* \mid 1^m \text{ where } m \text{ is odd}\}$$



$$L_3 = L_1 \circ L_2$$

In other words, the question means $L_3 = \{w \in \{0, 1\}^* \mid 0^n 1^m \text{ where } n \geq 0 \text{ and } m \text{ is odd}\}$

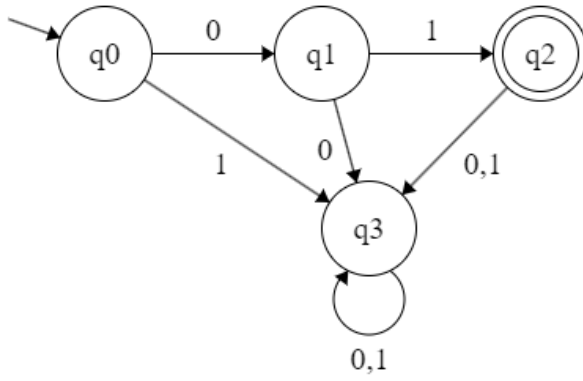


Note, there won't be a final state in q_0 .

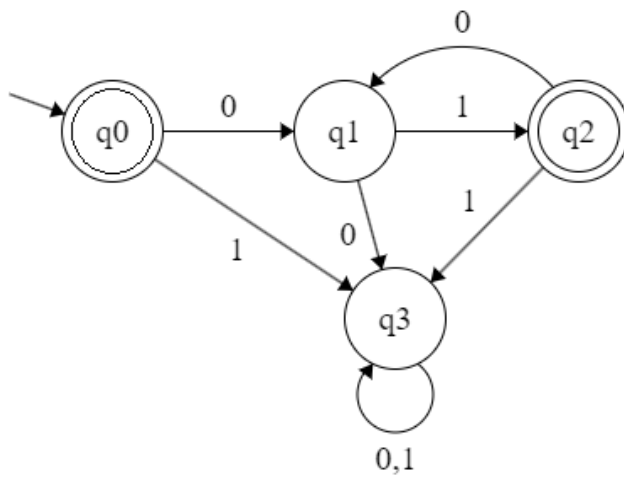
Lecture 4

Kleene star

$$L_4 = \{w \in \{0, 1\}^* \mid w \text{ is } 01\}$$



$$L_5 = L_4^*$$

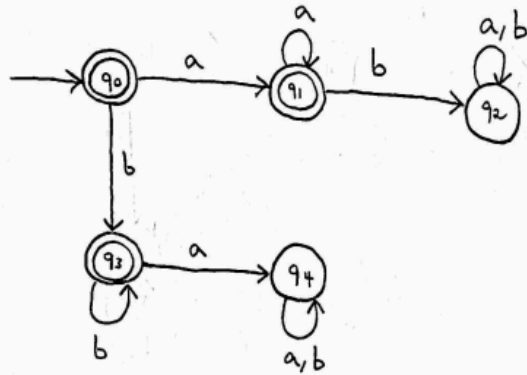


Lecture 4

$L_6 = \{w \in \{0, 1\}^* \mid w \text{ is any string not in } a^* \cup b^*\}$

$\Rightarrow \bar{L} = \{w \mid w \text{ is any string in } a^* \cup b^*\}$

$\Rightarrow$ DFA for $\bar{L}$:



$\Rightarrow$ DFA for L : Accept states in $\bar{L}$ become reject states in L
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